Breast Cancer Prediction Using Deep Learning

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***Abstract*—Breast cancer is one of the life-threatening diseases worldwide. Early diagnosis of breast cancer allows timely clinical intervention and targeted treatment that leads to an increase in the survival rate of patients. The existing methods for breast cancer detection mostly rely on classical methods, suffer from class imbalance problems and feature redundancy, which restricts their robustness and generalization capability. In this paper, breast cancer detection is implemented on the Wisconsin Breast Cancer Dataset. The class imbalance problem is mitigated using the Synthetic Minority Over-Sampling Technique (SMOTE). Redundant features are eliminated using Correlation Pruning. To lower computational cost and improve learning performance, dimensionality reduction is implemented using Principal Component Analysis (PCA) and Independent Component Analysis (ICA). Detection is implemented using machine learning algorithms such as Logistic Regression, Support Vector Machine, and Random Forest, along with deep learning models including Stacked Autoencoder (SAE), Denoising Autoencoder (DAE), Sparse Autoencoder, and Variational Autoencoder (VAE). The SAE coupled with a Deep Neural Network classifier outperformed all models, achieving a test accuracy of 98.83%.**

***Index Terms*—Breast cancer, deep learning, autoencoder, PCA, ICA, SMOTE, classification, ensemble learning**

I. Introduction

Breast cancer is a disease in which abnormal breast cells grow out of control and form tumours. Breast cancer cells begin inside the milk ducts and/or the milk-producing lobules of the breast. The earliest form (in situ) is not life-threatening and can be detected in early stages. However, cancer cells can spread into nearby breast tissue (invasion), creating tumours that cause lumps or thickening. Invasive cancers can further spread to nearby lymph nodes or other organs (metastasize), a process that can be life-threatening and fatal [1].

Breast cancer is still one of the most common and fatal cancers in the world. In 2022, around 2.3 million females were diagnosed with breast cancer, leading to approximately 670000 deaths globally [1]. Breast cancer is diagnosed in every country and at any age after puberty, with incidence increasing in later life. A major risk factor is gender: 99% of breast cancers are female, and 0.5–1% are male [1]. The national-level context exemplifies the scale of the problem: more than 3,300 Australians die from breast cancer annually — including forty men — and around 1 in 7 women and about 1 in 600 men are expected to be diagnosed with breast cancer during their lifetime. Most cases (around 80%) arise in women over the age of 50, but approximately 1,000 young women under-40 are diagnosed every year — about three a day. Even more alarming, incidence has skyrocketed over the past three decades, from roughly 9,800 new cases per year in 1994 to more than 20,000 new cases per year by 2025. Although the five-year relative survival rate is currently 93%, absolute diagnoses and deaths continue to increase [2], [3].

Due to this major public health burden of cancer, its gradual and accurate detection is extremely crucial and it can substantially aid in improving the prognosis and survival rates. Recently, computational methods based on machine learning and deep learning have proven their efficacy as a clinical aid in the diagnostic process, capable of producing objective, reproducible, and highly accurate classifications of breast tumors as benign or malignant [4], [5], [26]–[28].

In tackling these issues, this work introduces and tests a more complete analytical pipeline that combines advanced approaches to preprocessing, dimensionality reduction, class balancing and classification methods. The pipeline uses the Synthetic Minority Over-sampling Technique (SMOTE) to correct for class imbalance and produce a perfectly balanced training distribution [8]. To preprocess the feature space, we apply correlation based feature exclusion with a 0.95 threshold (i.e., $|r| > 0.95$) to remove redundant features and reduce multicollinearity [34], [37]. Our analysis systematically compares two related dimensionality reduction algorithms—Principal Component Analysis (PCA) while retaining 95% explained variance; and Independent Component Analysis (ICA)—to discover an optimal feature representation [9], [10].

This is done by evaluating a variety of classifiers under traditional machine learning algorithms—Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naïve Bayes,

Gradient Boosting—as well as ensemble methods which are Soft Voting and Stacking classifiers. In addition, this work explores the performance of four autoencoder-based deep neural network (DNN) architectures for unsupervised feature learning—namely: respectively Stacked Autoencoder (SAE), Denoising Autoencoder (DAE), Sparse Autoencoder (SpAE) and Variational Autoencoder (VAE) along with a downstream DNN classifier [11]–[13], [29]. All models undergo rigorous evaluation with 10-fold cross-validation on the balanced training set, and an independent evaluation from a held-out stratified test set covering 30% of data.

The principal contributions of this work are as follows:

- A systematic comparison of PCA and ICA dimensionality reduction methods over a series of classifier families.
- Dealing with dataset-dependent problems by combining SMOTE-based class balancing with correlation based feature pruning.
- A thorough benchmark of four autoencoder variants as feature extractors for downstream DNN classification.
- DAE+DNN architecture provides the best compromise between cross-validation peak consistency and generalization on the test set, and achieving 98.83% with SAE+DNN is valid for a single top performance metric label.

II. Methodology

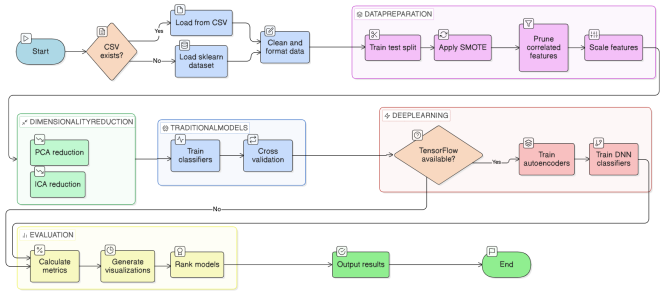


Fig. 1. Block diagram of the proposed breast cancer prediction pipeline, illustrating data loading, preprocessing, dimensionality reduction, classification, and evaluation stages.

A. Dataset Description

The target variable in the dataset is binary, indicating the presence of benign (B, encoded as 0) or malignant (M, encoded as 1) tumours. The dataset comprises of 357 benign samples (62.7%) and 212 malignant samples (37.3%), highlighting a significant class imbalance that may lead classifiers to favour the majority class [7]. Furthermore, the 30 features demonstrate considerable multicollinearity, with several feature pairs exhibiting correlations greater than 0.95. This collinearity can adversely affect model performance and elevate computational demands [6], [25].

B. Data Preprocessing

1) Train-Test Split: For pre-processing, the dataset undergoes a 70/30 train-test split using stratified sampling (random_state=40) to ensure class representation in both subsets.

2) Class Imbalance Handling with SMOTE: To address the class imbalance, SMOTE (Synthetic Minority Over-sampling Technique) is applied solely to the training set with parameters set to k_neighbors=3 and random_state=42. This process adjusts the distribution of samples, increasing the count from 250 benign and 148 malignant to a balanced 250 samples per class. In this context, each minority class sample (x_i) is augmented by generating synthetic instances based on its k nearest neighbours (x_{nn}):

$$x_{\text{new}} = x_i + \lambda \cdot (x_{nn} - x_i), \quad \lambda \in [0, 1] \tag{1}$$

Fig. 2 illustrates the balancing effect.

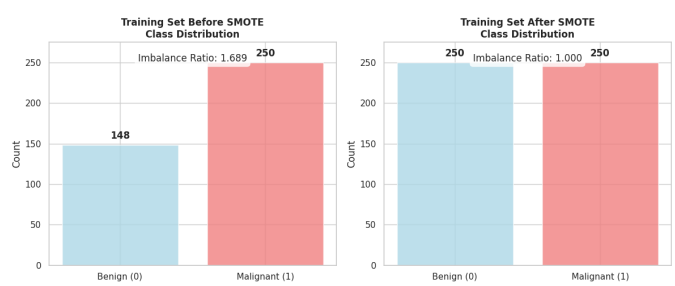


Fig. 2. Training set class distribution before and after applying SMOTE. The imbalance ratio improved from 1.689 to 1.000, with the minority class (Malignant) augmented from 148 to 250 samples.

3) Correlation-Based Feature Pruning: To address multicollinearity issues, the Pearson correlation coefficients are calculated for all feature pairs within the balanced training dataset. Features that display absolute pairwise correlations surpassing a threshold of $|r| > 0.95$ are pinpointed using the upper triangle of the correlation matrix, leading to the elimination of redundant features from both training and test datasets. This pruning process decreases dimensionality while maintaining important discriminative information.

Fig. 3 shows the correlation heatmap after pruning.

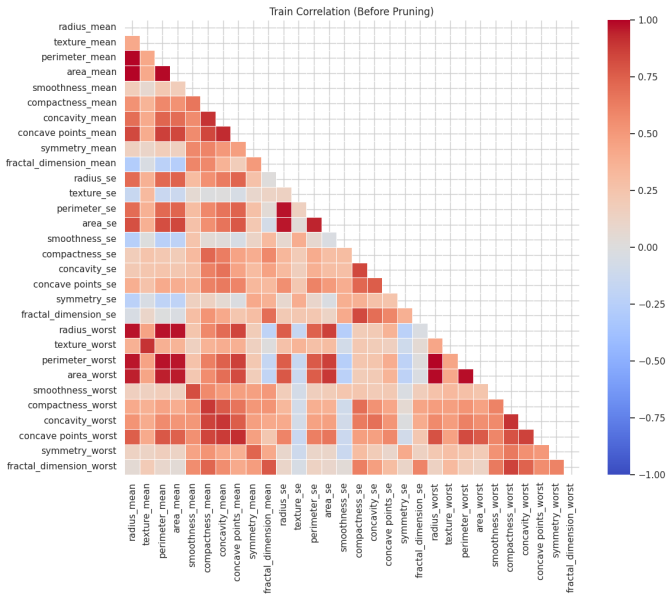


Fig. 3. Feature correlation heatmap after pruning highly correlated features (threshold $> |0.95|$).

C. Dimensionality Reduction

Two complementary dimensionality reduction techniques are employed:

1) Principal Component Analysis (PCA): Following standardization, PCA (n_components=0.95) is performed to retain components that account for 95% of the variance. The covariance matrix eigendecomposition:

$$C = \frac{1}{n-1} X^T X \tag{2}$$

$$C\mathbf{v} = \lambda \mathbf{v} \tag{3}$$

The explained variance ratio of the k-th component is:

$$\text{EVR}_k = \frac{\lambda_k}{\sum_{j=1}^{p} \lambda_j} \tag{4}$$

Fig. 4 and Fig. 5 show the scree plot and 2D class-separating projection.

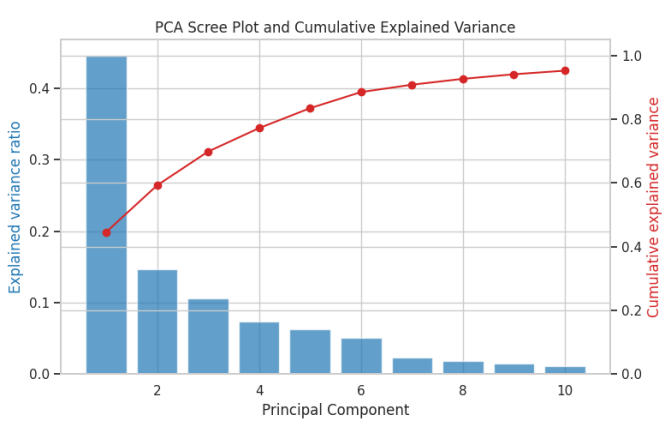


Fig. 4. PCA scree plot showing individual explained variance ratio (blue bars) and cumulative explained variance (red line) for each principal component.

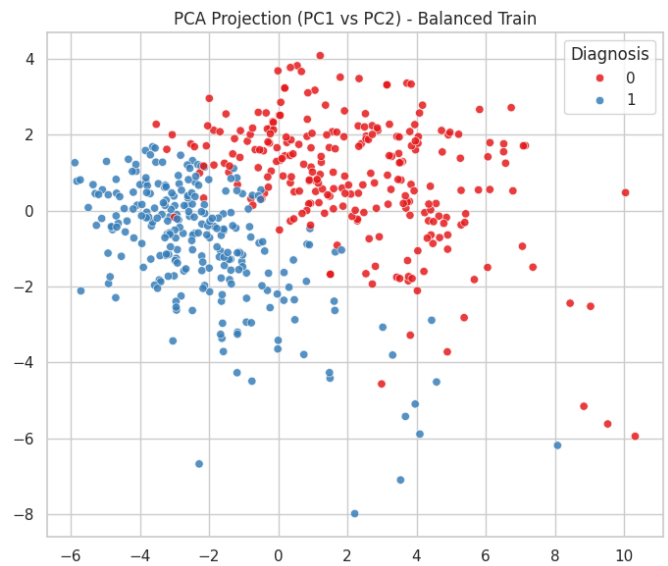


Fig. 5. PCA projection (PC1 vs PC2) of the balanced training set. Benign (0) shown in red and Malignant (1) in blue.

2) Independent Component Analysis (ICA): FastICA employs the same number of components as PCA (max_iter=1000, tol=1e-4) for a valid comparison:

$$X = A \cdot S, \quad S = W \cdot X \tag{5}$$

where A represents the mixing matrix, S denotes the independent sources, and $W = A^{-1}$ serves as the unmixing matrix.

D. Classification Models

1) Base Classifiers: Seven base classifiers are evaluated:

TABLE I
BASE CLASSIFIER CONFIGURATIONS

Classifier	Key Hyperparameters
Logistic Regression	max_iter=1000
Decision Tree	Default parameters
Random Forest	n_estimators=500, n_jobs=-1
KNN	Default (k=5)
SVM	kernel=rbf, gamma=scale, probability=True
Gaussian Naïve Bayes	Default parameters
Gradient Boosting	Default parameters

Logistic Regression [14] estimates the posterior probability using the sigmoid function and is optimized through binary cross-entropy loss (max_iter=1000). SVM [15] utilizes the radial basis function (RBF) kernel with γ=scale, transforming inputs into a higher-dimensional feature space where a maximum-margin hyperplane separates the classes. KNN [16] classifies new instances based on the majority vote amongst the $k = 5$ closest neighbours using Euclidean distance. Gaussian Naïve Bayes [17] presumes conditional independence of features and models each class-conditional distribution as Gaussian. Decision Tree [18] sequentially divides the feature space using Gini impurity as the criterion for splitting, while Random Forest [19] combines 500 independently trained trees

to decrease variance and enhance generalization. Gradient Boosting [20] incrementally fits weak learners to the residuals from the ensemble, minimizing a differentiable loss function. All base classifiers are implemented using scikit-learn [24].

2) Ensemble Methods: Two ensemble strategies are employed [35]:

- **Soft Voting Classifier:** Merges Logistic Regression, SVM, and Random Forest through soft voting with assigned weights of [1, 2, 2] respectively.
- **Stacking Classifier:** Takes SVM, Random Forest, and Gradient Boosting as base estimators, using Logistic Regression (max_iter=2000) as the meta-learner and incorporates 5-fold internal cross-validation.

The soft voting ensemble prediction is computed as:

$$\hat{y} = \arg\max_{c} \sum_{j=1}^{M} w_j \cdot P_j(y = c \mid x) \tag{6}$$

where w_j denotes the weight attributed to the j-th model.

The stacking meta-learner combines base learner predictions:

$$\hat{y} = f_{\text{meta}}\left(\hat{y}_1(x), \hat{y}_2(x), \ldots, \hat{y}_M(x)\right) \tag{7}$$

where $\hat{y}_j(x)$ represents cross-validated predictions from the base learners.

3) Autoencoder-Based DNN Classifiers: Four types of autoencoders are implemented using TensorFlow/Keras [21], [22], [32], each succeeded by a DNN classifier.

Stacked Autoencoder (SAE): This is a symmetric encoder-decoder with dimensions [128, 64, 32], utilizing ReLU activations, trained with Adam [31] (lr=1e-3) and MSE loss:

$$\mathcal{L}_{\text{recon}} = \frac{1}{N} \sum_{i=1}^{N} \|x_i - \hat{x}_i\|^2 \tag{8}$$

Denoising Autoencoder (DAE): Enhances the SAE by adding Gaussian noise (σ=0.1) at the input to learn features that are invariant to noise, reconstructing clean x from corrupted $\tilde{x}$:

$$\tilde{x} = x + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2) \tag{9}$$

The DAE reconstructs clean x from corrupted $\tilde{x}$ using loss (8).

Sparse Autoencoder (SpAE): Implements L1 activity regularization (λ=1e-4) on the bottleneck layer to encourage sparse activations:

$$\mathcal{L}_{\text{sparse}} = \mathcal{L}_{\text{recon}} + \lambda \sum_{j} |h_j| \tag{10}$$

where h_j represents hidden activations and λ is the sparsity coefficient.

Variational Autoencoder (VAE): Utilizes a probabilistic latent space (hidden=64, latent=16) incorporating reparameterization:

$$z = \mu + \sigma \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{11}$$

Reconstruction loss:

$$\mathcal{L}_{\text{recon}} = \sum_{j=1}^{d} (x_j - \hat{x}_j)^2 \tag{12}$$

KL divergence regularization:

$$D_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^{J} \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2\right) \tag{13}$$

Total VAE loss:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + D_{\text{KL}} \tag{14}$$

The deterministic z_mean is used for classification.

DNN Classifier Head: Each encoder is linked to Dense(32, ReLU) $\rightarrow$ Dropout [30] (0.2) $\rightarrow$ Dense(1, sigmoid), fine-tuned in an end-to-end manner using Adam and binary cross-entropy (with early stopping patience set to 12).

E. Pipeline Architecture

Each traditional/ensemble classifier is evaluated within two pipeline configurations:

- **PCA Pipeline:** StandardScaler $\rightarrow$ PCA (95% variance) $\rightarrow$ SMOTE (k=3) $\rightarrow$ Classifier
- **ICA Pipeline:** StandardScaler $\rightarrow$ FastICA (n=PCA components) $\rightarrow$ SMOTE (k=3) $\rightarrow$ Classifier

Autoencoder-based classifiers follow a distinct pipeline: StandardScaler $\rightarrow$ Autoencoder Pretraining (200 epochs, batch_size=32, EarlyStopping [33] patience=15) $\rightarrow$ DNN Classifier Training (200 epochs, batch_size=32, EarlyStopping [33] patience=12).

F. Evaluation Strategy

All models undergo evaluation via two complementary approaches. First, 10-fold cross-validation [23], [36] (KFold, shuffle=True, random_state=40) is executed on the balanced training dataset to gauge generalization accuracy and variability across data segments. Second, a held-out assessment on the original (unbalanced) 30% stratified test set evaluates real-world performance on unseen data. Standard classification metrics—Accuracy, Precision, Recall, and F1-Score—are reported along with confusion matrices. The Performance Gap, defined as the difference between mean CV accuracy and test accuracy, is calculated for each model to measure the risk of overfitting; smaller gaps suggest more dependable generalization.

III. Results

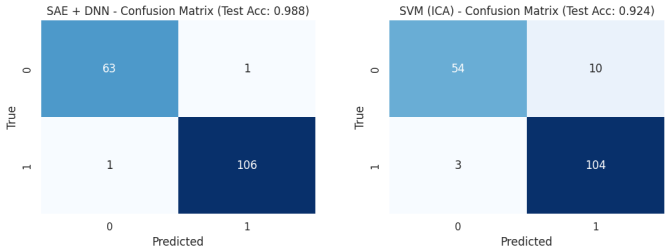
A. Model Performance Summary

Table II presents the complete performance comparison of all evaluated models, ranked by cross-validation accuracy.

TABLE II
PERFORMANCE COMPARISON OF ALL MODELS (RANKED BY CV ACCURACY)

Rank	Model	CV Acc	CV Std	Test Acc
1	DAE+DNN	0.9820	±0.0098	0.9766
2	SparseAE+DNN	0.9820	±0.0098	0.9649
3	SAE+DNN	0.9800	±0.0110	0.9883
4	SVM (ICA)	0.9760	±0.0250	0.9240
5	VotingSoft (PCA)	0.9760	±0.0265	0.9649
6	VotingSoft (ICA)	0.9760	±0.0265	0.9532
7	StackingLR (ICA)	0.9740	±0.0254	0.9357
8	SVM (PCA)	0.9720	±0.0256	0.9708
9	StackingLR (PCA)	0.9720	±0.0286	0.9649
10	LogReg (PCA)	0.9680	±0.0371	0.9649
11	LogReg (ICA)	0.9680	±0.0392	0.9649
12	KNN (PCA)	0.9620	±0.0328	0.9474
13	RF (PCA)	0.9540	±0.0400	0.9532
14	GradBoost (ICA)	0.9540	±0.0310	0.9181
15	RF (ICA)	0.9520	±0.0337	0.9415
16	GradBoost (PCA)	0.9520	±0.0349	0.9357
17	NaiveBayes (PCA)	0.9500	±0.0205	0.9532
18	KNN (ICA)	0.9420	±0.0227	0.9064
19	NaiveBayes (ICA)	0.9360	±0.0367	0.9415
20	DecTree (PCA)	0.9220	±0.0477	0.9064
21	DecTree (ICA)	0.9180	±0.0533	0.8772

Fig. 6 compares the confusion matrices of the best-performing deep learning model (SAE + DNN) and a representative classical model (SVM with ICA).



(a) SAE + DNN (Acc: 98.8%) (b) SVM with ICA (Acc: 92.4%)

Fig. 6. Confusion matrices comparing SAE + DNN with SVM (ICA). The SAE + DNN achieved only 2 misclassifications compared to 13 for SVM (ICA).

B. Top 6 Models Analysis

TABLE III
DETAILED ANALYSIS OF TOP 6 MODELS

Rank	Model	CV Acc	Test Acc	Result
1	DAE+DNN	0.9820±0.0098	0.9766	Excellent
2	SparseAE+DNN	0.9820±0.0098	0.9649	Excellent
3	SAE+DNN	0.9800±0.0110	0.9883	Excellent
4	SVM (ICA)	0.9760±0.0250	0.9240	Caution
5	VotingSoft (PCA)	0.9760±0.0265	0.9649	Excellent
6	VotingSoft (ICA)	0.9760±0.0265	0.9532	Good

Fig. 7 shows the top 6 models ranked by cross-validation accuracy, along with a comparison of CV and test accuracy.

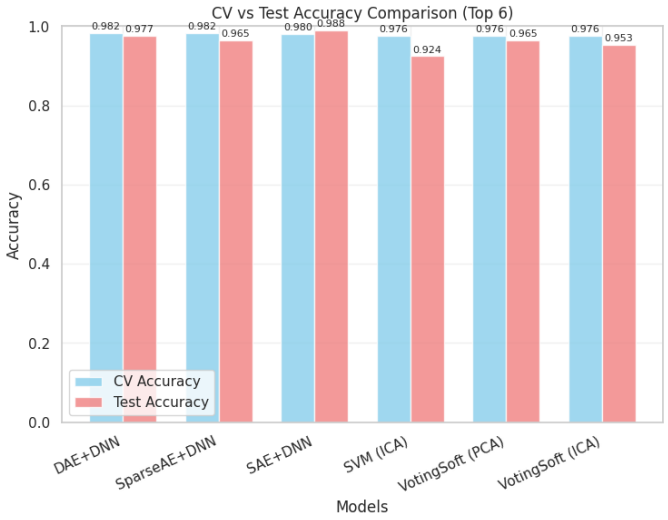


Fig. 7. Top 6 models ranked by cross-validation accuracy (left) and CV vs Test accuracy comparison (right).

C. Key Findings

By CV accuracy, autoencoder-based models took the top three spots, with DAE, SpAE, and SAE all greater than 98% and standard deviations $\leq$0.011 — indicating consistent, stable performance across folds. Among them, SAE+DNN showed the highest test accuracy of 98.83%, even though ranking third in CV. Its 0.83% performance gap indicates strong generalization. On the other hand, DAE+DNN, achieved the best overall consistency. It topped CV accuracy at 98.20% while maintaining the smallest gap of just 0.54%, making it the most reliable choice end-to-end.

PCA-based pipelines outperformed ICA by roughly 2 percentage points on average (94.81% vs. 92.81%), with the gap most visible in SVM (97.08% vs. 92.40%) and KNN (94.74% vs. 90.64%). SVM with ICA also showed clear signs of overfitting, with a 5.20% drop between CV and test accuracy — the biggest among all top models.

Ensemble methods were competitive, with Soft Voting (PCA) reaching 96.49% test accuracy and a tight 1.11% gap. The weakest performer was Decision Tree regardless of the dimensionality reduction method used.

IV. CONCLUSION

This study conducted a comprehensive evaluation of machine learning, ensemble, and deep learning approaches for breast cancer classification on the WDBC dataset. We developed a pipeline to handle class imbalance, feature redundancy, and high dimensionality by testing 21 model setups, including seven traditional classifiers, two ensemble methods, and four autoencoder-based deep learning models. The autoencoder-based feature learning was very effective, with the DAE+DNN model recommended as the best option. It achieved a cross-validated accuracy of 98.20% (± 0.98%) and a test accuracy of 97.66%, with a slight performance gap of 0.54%. Its noise-injection during pretraining helps it extract strong features that withstand input changes. The SAE+DNN model had the

highest test accuracy at 98.83%, indicating that unsupervised pretraining yields better features than manual methods. Additionally, PCA-based pipelines outperformed ICA by about 2 percentage points, preserving more relevant information. SMOTE and correlation pruning are crucial for fixing class imbalance and feature redundancy. Ensemble methods like Soft Voting with PCA (96.49% test accuracy) are good alternatives when deep learning tools are unavailable.

A. Limitations and Future Work

Acknowledging the inherent limitations of this study is crucial. All evaluations were carried out on a single dataset (WDBC); it is imperative to validate findings on larger, multi-institutional datasets to enhance the generalizability of the results. The process of hyperparameter optimization has been confined to manual selection; however, implementing systematic tuning through Bayesian optimization may yield further improvements in model performance. Additionally, the autoencoder architectures employed fixed encoding dimensions; conducting a comprehensive neural architecture search could potentially identify more optimal configurations for improved outcomes. Future research avenues should focus on extending the framework to encompass multi-class cancer classification, integrating additional imaging modalities for a more holistic approach, and deploying the most effective models as clinical decision support tools to assist healthcare professionals in real-world settings.

Acknowledgment

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